

So far we have only considered series of positive term. If we want we could analyze series with both positive and negative terms. Specifically, we look at alternating series.

Def

An alternating series is a series whose terms alternate between positive and negative. That is, if we have the series

$$a_1 - a_2 + a_3 - a_4 + a_5 - a_6 + \dots$$

We usually write these with positive terms $b_n = |a_n|$ and $(-1)^n$ or $(-1)^{n-1}$ to change the sign.

For example:

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n}$$
$$-\frac{1}{2} + \frac{2}{3} - \frac{3}{4} + \frac{4}{5} - \dots = \sum_{n=1}^{\infty} (-1)^n \frac{n}{n+1}$$

To evaluate the convergence of Alternating Series
we use the following test:

Alternating Series Test

If the alternating Series

$$\sum_{n=1}^{\infty} (-1)^{n-1} b_n = b_1 - b_2 + b_3 - b_4 + b_5 - b_6 + \dots$$

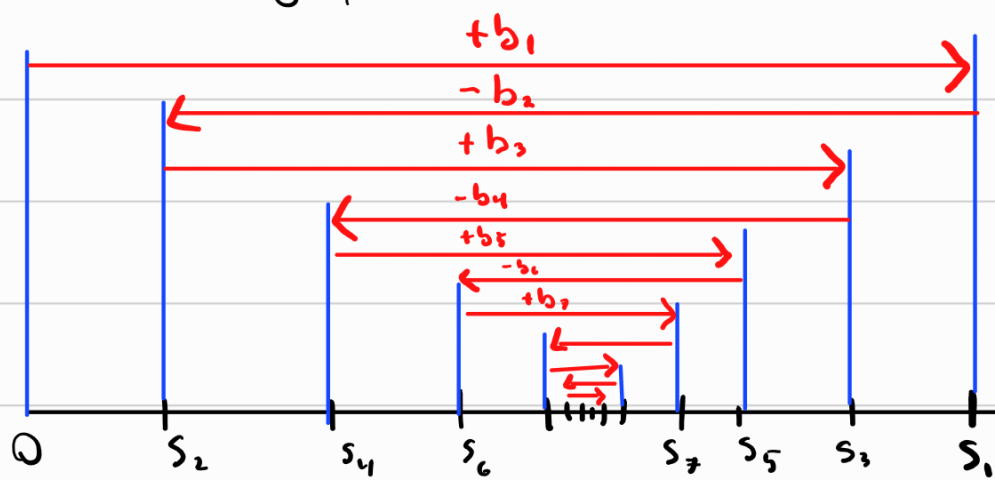
where $b_n > 0$ is such that

1) $b_{n+1} \leq b_n$ for all n (i.e. $\{b_n\}$ is decreasing)

2) $\lim_{n \rightarrow \infty} b_n = 0$

then $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$ is Convergent.

Consider the following picture



Since each $b_{n+1} \leq b_n$, when we subtract $S_{n+1} = S_n - b_{n+1}$ we can't pass the previous sum, S_2 we slowly move to a central point.

Pf/ Consider the even partial sums

$$S_2 = b_1 - b_2 \geq 0 \quad \text{as } b_1 \geq b_2$$

$$S_4 = S_2 + (b_3 - b_4) \geq S_2 \quad \text{as } b_3 \geq b_4$$

in general

$$S_{2n} = S_{2n-2} + (b_{2n-1} - b_{2n}) \geq S_{2n-2} \quad \text{as } b_{2n-1} \geq b_{2n}$$

So

$$0 \leq S_2 \leq S_4 \leq \dots \leq S_{2n} \leq \dots$$

We also see that

$$S_{2n} = b_1 - b_2 + b_3 - b_4 + b_5 - \dots + (b_{2n-2} - b_{2n-1}) - b_{2n}$$

$$S_{2n} = b_1 - (b_2 - b_3) - (b_4 - b_5) - \dots - (b_{2n-2} - b_{2n-1}) - b_{2n}$$

$\uparrow$
 $b_{i-1} \geq b_i$

So

$$S_{2n} \leq b_1$$

Thus $\{S_{2n}\}$ is increasing and bounded,
and by monotone convergence Thm, converges.

$$\text{Say } \lim_{n \rightarrow \infty} S_{2n} = S$$

Now for odd partial sums

$$S_{2n+1} = S_{2n} + b_{2n+1}$$

$$\begin{aligned} \lim_{n \rightarrow \infty} S_{2n+1} &= \lim_{n \rightarrow \infty} S_{2n} + \lim_{n \rightarrow \infty} b_{2n+1} \\ &= S + 0 \\ &= S \end{aligned}$$

So we then see that

$$\sum_{n=1}^{\infty} (-1)^{n-1} b_n = \lim_{n \rightarrow \infty} S_n = S \quad \text{converges}$$

Ex: Alternating harmonic Series

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$$

We notice $b_{n+1} < b_n$

$$\frac{1}{n+1} < \frac{1}{n}$$

and

$$\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\text{So } \sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n} \text{ Converges}$$

Ex:

$$\sum_{n=1}^{\infty} (-1)^n \frac{3n}{4n-1}$$

$$\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \frac{3n}{4n-1} = \frac{3}{4} \neq 0$$

So we can't use
Alternating Series test

We notice

$$\lim_{n \rightarrow \infty} (-1)^n \frac{3n}{4n-1} = \frac{3}{4} \lim_{n \rightarrow \infty} (-1)^n \quad \text{DNE}$$

So Series diverges

Ex:
$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n^2}{n^3+1}$$

This looks like alternating harmonic, so we think it converges.

Let's check

$$\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \frac{n^2}{n^3+1} = \lim_{n \rightarrow \infty} \frac{\frac{1}{n}}{1+\frac{1}{n^3}} = 0$$

To check decreasing

$$f(x) = \frac{x^2}{x^3+1}, \quad f'(x) = \frac{x(2-x^2)}{(x^3+1)^2}$$

$$\text{and } f'(x) < 0 \text{ for } x > \sqrt[3]{2}$$

As always, we can ignore a finite amount of terms, so eventually decreasing is fine.

So, b_n is decreasing eventually and $\lim_{n \rightarrow \infty} b_n = 0$

$$\text{So } \sum_{n=1}^{\infty} (-1)^{n+1} \frac{n^2}{n^3+1} \text{ Converges}$$

Absolute Convergence

Given any Series $\sum a_n$ we can consider the corresponding Series

$$\sum_{n=1}^{\infty} |a_n| = |a_1| + |a_2| + |a_3| + \dots$$

If this Series converges, the original Series has some special Properties.

Def: A Series $\sum a_n$ is called absolutely Convergent if the Series $\sum |a_n|$ is Convergent.

Note: Any Series of Positive terms that Converges is absolutely Convergent inherently

Ex:

The alternating Series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2} = 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$$

is absolutely Convergent as

$$\sum_{n=1}^{\infty} \left| \frac{(-1)^{n-1}}{n^2} \right| = \sum \frac{1}{n^2} \text{ Converges}$$

Def: A series $\sum a_n$ is called conditionally convergent if it is convergent but not absolutely convergent.

That is, if $\sum a_n$ converges but $\sum |a_n|$ diverges.

Ex:

$\sum \frac{(-1)^{n-1}}{n}$ converges but

$$\sum \left| \frac{(-1)^{n-1}}{n} \right| = \sum \frac{1}{n} \text{ diverges}$$

So Harmonic Series is Conditionally Convergent

Thm:

If a series $\sum a_n$ is absolutely convergent,
it is convergent

PP/ Notice $|a+b| \leq |a|+|b|$

So,

$$0 \leq \left| \sum a_n \right| \leq \sum |a_n|$$

if the latter is convergent so is $\sum a_n$.

Ex: $\sum \frac{\cos(n)}{n^2}$

$$0 \leq \frac{|\cos(n)|}{n^2} \leq \frac{1}{n^2}$$

$$\sum \frac{1}{n^2} \text{ converges, so } \sum \frac{|\cos(n)|}{n^2} \text{ converges}$$

and so does

$$\sum \frac{\cos(n)}{n^2} \text{ converges}$$

Ex: $\sum \frac{(-1)^n}{n^3}$ is absolutely convergent as $\sum \frac{1}{n^3}$ converges

$$\sum \frac{(-1)^n}{\sqrt{n}} \text{ is conditionally convergent as } \sum \frac{1}{\sqrt{n}} \text{ diverges}$$

but $\sum \frac{(-1)^n}{\sqrt{n}}$ converges (alternating series test)

$$\sum (-1)^n \frac{n}{2n+1} \text{ diverges by divergence test.}$$